iDABET 產學研工程人才實務能力卓越基地計畫
DEVELOPMENT APPLICATION BASE OF ENGINEERING TALENTS

The Development Application Base of Engineering Talents (iDABET)
Project of the Industrial Development Administration, Ministry of Economic Affairs

2025 Practical Ability Development Implementation Unit Project – Engineering Talent Practical Ability Development Record Form

Project No. of the Practical Ability Development Implementation Unit: DABET11403
No. of Engineering Talent: 40
Name of the Practical Ability Development Implementation Unit: 財團法人金屬工業研究發展中心/光電技術組
Project Name of the Practical Ability Development Implementation Unit: 半導體產業之真空設備與製程能力發展與人工智慧控制應用
Name of the Engineering Talent: 蘇翼

Item	Date	Instructor	No. of Hours (Note 1)	Type of Practical Ability Development Topic (Note 2)	Topic	Description
1	5/2	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Setting up parameters for the Indoor positioning system using 4 beacons.
2	5/5	李宗信	5	1	Indoor Positioning System Research (Thesis)	Independent Study: Testing the parameters for the Indoor positioning system using 4 beacons.
3	5/7	陳泰宏	2	2	CVD Applications	Visited Functional Coating System company for an insight of how CVD is applied to products.
4	5/8	吳以德	2	2	PECVD Systems	Principles of using Plasma to enhance CVD.
5	5/8	林家豪/李宗信	2	1	Application of PECVD on samples	Hands-on practice of the PECVD process.
6	5/9	李宗信	4	1	Indoor Positioning System Research (Thesis)	Going over the goals and expectations of Indoor Positioning.
7	5/10	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Programming an interface to analyze and collect position data.
8	5/13	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Programming an interface to analyze and collect position data continues (continued).

9	5/14	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Debugging the interface for analysis and position data collection(continued).
10	5/15	李宗信	2	1	Indoor Positioning System/ Semiconductor X Drone Application	Teams online meet discussing Semiconductor X Drone Application in Industrial Use.
11	5/16	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Debugging the interface for analysis and position data collection.
12	5/17	李宗信	7	1	Indoor Positioning System Research (Thesis)	Independent Study: Debugging the interface for analysis and position data collection (continued).
13	5/19	李宗信	6	1	Indoor Positioning System Research (Thesis)	Independent Study: Making changes to the A, B, C, and D beacon positions.
14	5/20	李宗信	6	1	Indoor Positioning System Research (Thesis)	Independent Study: Testing the accuracy of the system.
15	5/22	李宗信	2	2	"Besides Tech, What else is lacking?" Research insight	Learning about the necessary skills needed to have and to be aware of will on the research journey.
16	5/22	黃秉敦	2	1	PECVD / ALD Coating Technology	Review of PECVD and an Introduction to ALD
17	5/23	李宗信	4	1	Indoor Positioning System Research (Thesis)	Independent Study: Testing the accuracy of the system (continued).
Total Hours on Practical Ability Development Topics			Total _____64___ hours.			

Please add rows to the table as needed.

Notes: (1) The hours for implementing practical ability development topics should be filled in according to the actual implementation status between March 1 and August 31.

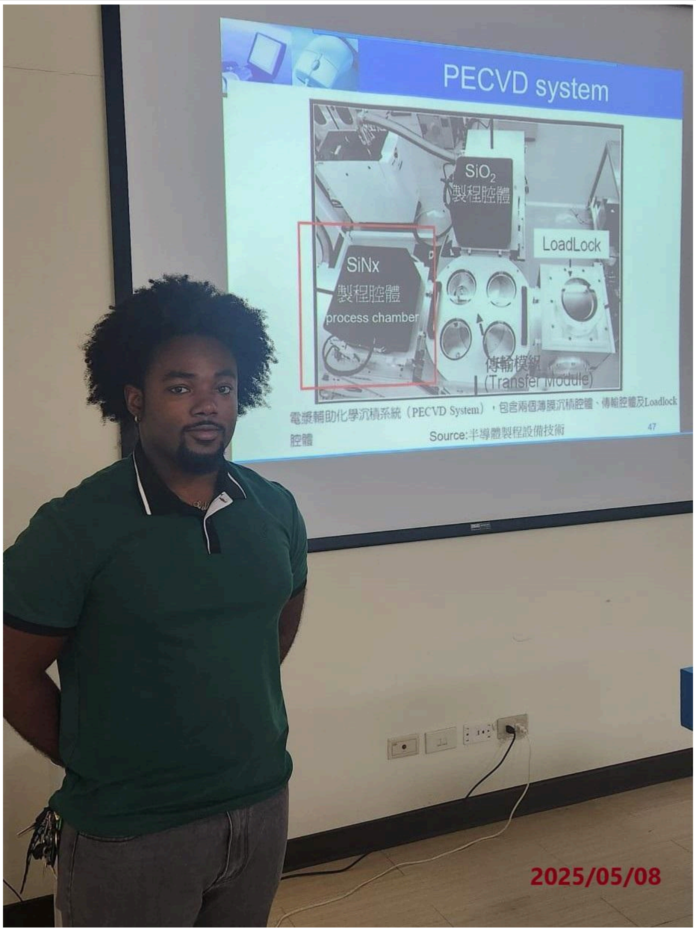
(2) Type of practical ability development topics: 1. Practice-based; 2. Industry collaboration-based.

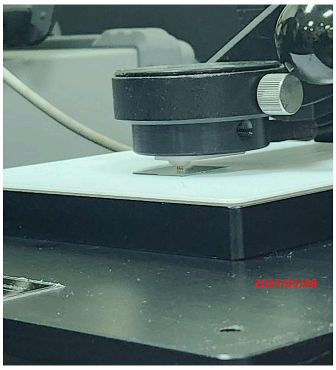
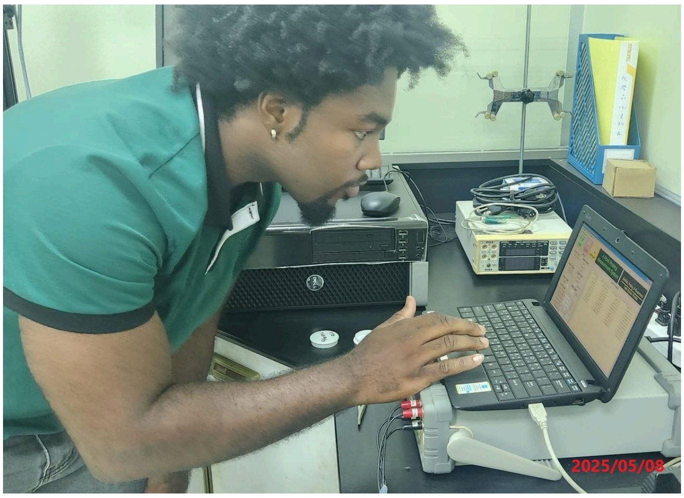
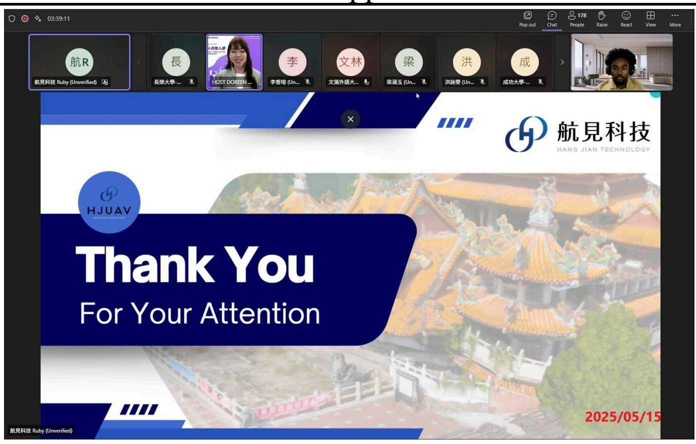
(3) Please attach at least 6 dated photos (yyyy/mm/dd) as evidence of the engineering talent's participation in practical ability development topics or activities.

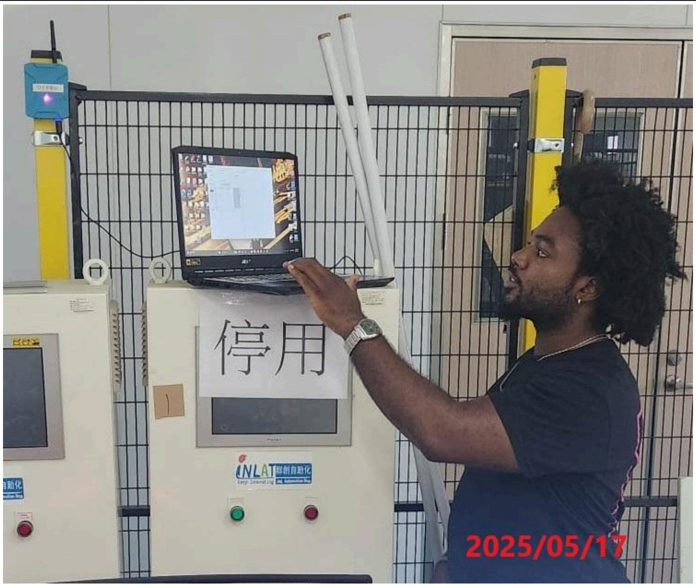
(4) Both the instructor and engineering talent shall sign at the bottom of the record form.

(5) To implement the "2025 Development Application Base of Engineering Talents (iDABET) Project", the Industrial Development Administration, Ministry of Economic Affairs will collect, process and use the personal data of the party involved without notifying the said party, as prescribed in Subparagraph 2 of Paragraph 2 of Article 8 of the Personal Data Protection Act.

【Photos of Participation in Practical Ability Development Topics or Activities】

 方均科技股份有限公司 Company tour	Today, we visited the Functional Coating System (方均科技股份有限公司), where we learned about their parylene nano coating process. This specialized coating offers multi-functional protection for electronic products, including IPX8 waterproofing, resistance to salt spray, chemical corrosion (acid and alkali), and high-voltage breakdown, all while maintaining signal continuity.
	Plasma-Enhanced Chemical Vapor Deposition (PECVD) is a process used to deposit thin films from a gas state (vapor) to a solid state on a substrate, using plasma to enhance the chemical reaction. This technique is widely used in semiconductor manufacturing, solar cells, MEMS, and optical coatings.

PECVD Class lecture	
   PECVD Application	Here, we got to check for the uniformity of the thin film. Only under the right parameters, will we get accurate reading and measurement.
 Semiconductor × Robotics & Drone Applications	This was a very interesting online meeting where the speakers introduced Semiconductor x Robotics and drone application in the real world. They discussed the pros and cons of their application.

 Indoor Positioning System Research (Thesis)	I'm currently debugging the interface for analysis and position data collection. To optimize accuracy and precision, the beacons should be placed in areas with minimal noise or interference.
 Atomic Layer Deposition	This machine was previously capable of performing Atomic Layer Deposition (ALD); however, the necessary components are currently under repair. As a result, we were only able to perform the Plasma-Enhanced Chemical Vapor Deposition (PECVD) method on the glass surface. To evaluate the effectiveness of the thin film, a drop of water was placed on the glass sample before and after coating. The contact angle of the water droplet significantly decreased, indicating a change in the surface properties.

Notes:

1. Please add rows to the table as needed. Each photo shall be dated at the bottom right, and each photo shall be accompanied by descriptions below it to explain the topic.
2. Please provide the original files of 1 to 3 photos (JPEG files, each above 1MB in size and with a resolution above 300 DPI).